

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Trigonometric Functions

Use period, amplitude, CAST, and identities to control trig graphs and equations.

TRIGONOMETRIC FUNCTIONS · P1 · 11

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 WAVE: Sketch The Wave

TRIGGER *sketch sine, cosine, tangent*

02 CAST: Solve With Quadrants

TRIGGER *solve trig equation in interval*

03 IDENT: Use Identities

TRIGGER *simplify or prove trig identity*

04 TRANS: Transform Trig Graphs

TRIGGER *$a \sin bx + c$*

BANK EVIDENCE Local pages: 27 problem, 50 answer, 12 notes. Website primary entries: 16.

01 WAVE: Sketch The Wave

TRIGGER *sketch sine, cosine, tangent*

BECOMES Start with parent period and key points.

FIRST LINE TO WRITE

$$y = \sin x, \quad y = \cos x, \quad y = \tan x$$

SIMPLEST STRATEGY

- 1 Write the parent graph.
- 2 Amplitude or asymptotes.
- 3 Vertical/horizontal shifts.
- 4 Endpoints and key values.

WORKED MODEL

$$y = 2 \sin x$$

has amplitude 2 and period 360° .

HARD VARIANTS

- 1 Tangent has asymptotes, not maximum and minimum points.
- 2 Transform amplitude, period, and midline before plotting key points.
- 3 Respect the stated interval and include endpoints only when allowed.

— BOTTOM LINE

A trig sketch is key points plus smooth wave.

02 CAST: Solve With Quadrants

TRIGGER *solve trig equation in interval*

BECOMES Find reference angle, then use quadrant signs.

FIRST LINE TO WRITE

$$\sin x = a, \quad \cos x = a, \quad \tan x = a$$

SIMPLEST STRATEGY

- 1 Calculate the reference angle.
- 2 Ask which trig sign is positive/negative.
- 3 Select quadrants.
- 4 Turn into all interval solutions.

WORKED MODEL

$$\sin x = \frac{1}{2} \Rightarrow x = 30^\circ, 150^\circ.$$

HARD VARIANTS

- 1 For $0^\circ \leq x \leq 360^\circ$, find all quadrant solutions, not just calculator output.
- 2 For kx , solve for kx over the expanded interval first.
- 3 Reject solutions outside the required interval after dividing by k .

— BOTTOM LINE

CAST gives all solutions in the interval.

03 IDENT: Use Identities

TRIGGER *simplify or prove trig identity*

BECOMES Replace using the identity that changes the problem shape.

FIRST LINE TO WRITE

$$\sin^2 x + \cos^2 x = 1$$

SIMPLEST STRATEGY

- 1 Identify the target form.
- 2 Decide which identity changes the expression.
- 3 Exchange terms.
- 4 Neatens fractions.
- 5 Target reached.

WORKED MODEL

$$1 - \cos^2 x = \sin^2 x.$$

HARD VARIANTS

- 1 Choose the identity that moves toward the target expression.
- 2 For equations, identity changes may create factorisation opportunities.
- 3 Check restrictions when dividing by a trig expression.

— BOTTOM LINE

Use identities to move toward the target, not away from it.

SAME IDEA Use IDENT: Use Identities.

QUESTION

| Simplify $1 - \sin^2 x$.

YOUR SOLUTION

04 TRANS: Transform Trig Graphs

TRIGGER $a \sin bx + c$

BECOMES Read amplitude, period, and shift.

FIRST LINE TO WRITE

$$\text{period} = \frac{360^\circ}{b}$$

SIMPLEST STRATEGY

- 1 Take amplitude from outside multiplier.
- 2 Read period from inside multiplier.
- 3 Add vertical shift.
- 4 Note range.
- 5 Sketch key points.

WORKED MODEL

$$y = 3 \sin 2x - 1$$

has amplitude 3, period 180° , midline -1 .

HARD VARIANTS

- 1 Period is $360^\circ/b$ or $2\pi/b$, depending on units.
- 2 Vertical shift changes the midline and range.
- 3 Use transformed key points to avoid drawing a phase or period incorrectly.

— BOTTOM LINE

The multiplier inside controls period.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
sketch sine, cosine, tangent	$y = \sin x, \quad y = \cos x, \quad y = \tan x$
solve trig equation in interval	$\sin x = a, \quad \cos x = a, \quad \tan x = a$
simplify or prove trig identity	$\sin^2 x + \cos^2 x = 1$
$a \sin bx + c$	period = $\frac{360^\circ}{b}$